

Hypertension (High Blood Pressure)

1. Overview and Definition

Hypertension is a condition in which the force of blood against the artery walls is persistently too high. Blood pressure is expressed as two numbers: systolic (pressure when heart beats) over diastolic (pressure when heart rests). Normal blood pressure is below 120/80 mmHg.

Hypertension is defined when readings consistently exceed 130/80 mmHg (AHA 2017 guidelines) or 140/90 mmHg (older WHO/Indian guidelines). It is classified as: Elevated (120-129/<80), Stage 1 (130-139/80-89), Stage 2 ($\geq 140/\geq 90$), and Hypertensive Crisis ($>180/>120$).

Hypertension affects over 1.28 billion adults worldwide and is responsible for an estimated 10.4 million deaths per year. In India, approximately 28-29% of adults are hypertensive, and urban prevalence can reach 33%. Despite its prevalence, nearly half of affected people are unaware of their condition.

The condition is called the 'silent killer' because most people experience no symptoms even when blood pressure reaches dangerously high levels. This makes regular screening critically important, especially for those with risk factors.

2. Pathophysiology

How Blood Pressure Is Regulated

Blood pressure is maintained through a complex interplay of cardiac output, systemic vascular resistance, blood volume, and neurohumoral factors. The renin-angiotensin-aldosterone system (RAAS) is a key regulator — when blood pressure drops, kidneys release renin, which ultimately produces angiotensin II, a potent vasoconstrictor.

The sympathetic nervous system also plays a major role. Stress, physical activity, and emotional arousal activate the sympathetic nervous system, raising heart rate and vascular resistance. Chronic activation can permanently elevate blood pressure.

Baroreceptors in the aortic arch and carotid sinuses provide short-term feedback. Kidneys control long-term blood pressure by regulating sodium and water retention. Any disruption in

these systems can result in sustained hypertension.

Types of Hypertension

Primary (essential) hypertension accounts for 90-95% of all cases. It has no identifiable single cause but is associated with genetic predisposition, aging, obesity, high-sodium diet, sedentary lifestyle, and excessive alcohol intake.

Secondary hypertension (5-10%) has an identifiable underlying cause including: chronic kidney disease, primary aldosteronism, renovascular disease (renal artery stenosis), obstructive sleep apnea, thyroid disorders, pheochromocytoma, coarctation of the aorta, and certain medications (NSAIDs, oral contraceptives, decongestants, stimulants).

Isolated systolic hypertension (high systolic with normal diastolic) is common in older adults due to arterial stiffening. Masked hypertension is normal in clinic but high at home. White coat hypertension is the reverse — high in clinic, normal elsewhere.

3. Risk Factors

Non-modifiable risk factors include: age (risk increases significantly after 55 in men and 65 in women), family history of hypertension, race (higher prevalence in South Asian and African-descent populations), and chronic kidney disease.

Modifiable risk factors include: obesity (BMI >30), physical inactivity, high-sodium diet (>5g salt/day), low potassium intake, excessive alcohol consumption (>2 drinks/day), smoking, chronic stress, sleep disorders, and use of certain medications.

In India, rapid urbanization has increased hypertension risk through dietary changes (increased processed food consumption), sedentary jobs, stress, and reduced physical activity. Rural-urban migration is a significant contributing factor.

4. Symptoms and Clinical Presentation

Most patients with hypertension are completely asymptomatic. This is precisely why routine blood pressure checks are important, as the first presentation can sometimes be a hypertensive crisis, heart attack, or stroke.

When symptoms do occur, they may include: morning headaches (occipital region), nosebleeds (epistaxis), tinnitus (ringing in ears), visual disturbances or blurred vision, dizziness, palpitations, and fatigue.

Hypertensive urgency is severely elevated blood pressure (>180/120) WITHOUT end-organ damage. Symptoms may include severe headache and anxiety but no neurological deficits or chest pain.

Hypertensive emergency is severely elevated blood pressure WITH end-organ damage. This includes hypertensive encephalopathy (confusion, seizures), acute MI, unstable angina, aortic dissection, acute pulmonary edema, acute renal failure, or eclampsia in pregnancy. This is a life-threatening emergency requiring immediate hospitalization.

5. Diagnosis and Monitoring

Measurement Standards

Blood pressure should be measured with a validated device after 5 minutes of rest, seated with back supported, feet flat on floor, and arm at heart level. Two measurements 1-2 minutes apart should be taken, and the average recorded. Measurements should be taken in both arms initially.

Home blood pressure monitoring (HBPM) is recommended to confirm diagnosis and assess treatment. Morning (before medications) and evening readings over 7 days provide the most reliable average. Target for home readings is <135/85 mmHg.

Ambulatory blood pressure monitoring (ABPM) — worn for 24 hours — is the gold standard for diagnosing white coat and masked hypertension. It also provides nocturnal readings; a lack of nighttime dipping (non-dipper pattern) is associated with higher cardiovascular risk.

Investigations

Initial workup includes: urinalysis (protein, blood, glucose), serum creatinine and electrolytes, fasting blood glucose, fasting lipid profile, complete blood count, ECG, and fundoscopy.

Additional tests for suspected secondary hypertension: renal ultrasound, plasma renin and aldosterone, 24-hour urine catecholamines (for pheochromocytoma), thyroid function tests, sleep study for sleep apnea.

Target organ damage assessment: echocardiogram for LVH, urine albumin-to-creatinine ratio for renal involvement, carotid intima-media thickness (CIMT), and ankle-brachial index (ABI).

6. Treatment — Lifestyle Modifications

The DASH diet (Dietary Approaches to Stop Hypertension) is the most evidence-based dietary intervention. It emphasizes fruits, vegetables, whole grains, lean proteins, and low-fat dairy while limiting saturated fats, red meat, sweets, and sodium. It can reduce systolic BP by 8-14 mmHg.

Sodium restriction: limit dietary sodium to less than 2,300 mg/day (ideally 1,500 mg for most adults). This means reducing processed foods, pickles, papad, salty snacks, and restaurant meals. Even a moderate reduction of 1,000 mg/day can significantly lower blood pressure.

Regular aerobic exercise — at least 150 minutes of moderate-intensity activity per week (walking, swimming, cycling) — can reduce systolic BP by 5-8 mmHg. Isometric resistance exercise (wall squats, handgrip) has also shown meaningful BP reduction.

Weight loss is one of the most effective lifestyle interventions — each kilogram of weight loss reduces systolic BP by approximately 1 mmHg. Achieving a healthy waist circumference (men <90 cm, women <80 cm for South Asians) is the target.

Limit alcohol to no more than 1 drink/day for women and 2 drinks/day for men. Excessive alcohol is responsible for approximately 16% of hypertension cases worldwide. Quitting smoking does not directly lower blood pressure but dramatically reduces overall cardiovascular risk.

Stress management through mindfulness meditation, yoga, deep breathing (pranayama), and adequate sleep (7-8 hours) can help lower blood pressure. Chronic psychological stress activates the sympathetic nervous system and RAAS system.

7. Pharmacological Treatment

First-Line Medications

ACE inhibitors (enalapril 5-40 mg, lisinopril 10-40 mg, ramipril 2.5-10 mg): block conversion of angiotensin I to angiotensin II, reducing vasoconstriction and aldosterone secretion. Side effect: dry cough (10-15% of patients), angioedema (rare but serious). Contraindicated in

pregnancy.

Angiotensin Receptor Blockers — ARBs (losartan 50-100 mg, telmisartan 40-80 mg, valsartan 80-320 mg): block angiotensin II receptors directly. Similar efficacy to ACE inhibitors but without the cough. Preferred when ACE inhibitor cough is problematic. Also contraindicated in pregnancy.

Calcium Channel Blockers — CCBs (amlodipine 5-10 mg, felodipine 5-10 mg, nifedipine SR 30-60 mg): relax arterial smooth muscle, reducing vascular resistance. Dihydropyridine CCBs (amlodipine) are preferred for hypertension; non-dihydropyridines (verapamil, diltiazem) also reduce heart rate.

Thiazide Diuretics (hydrochlorothiazide 12.5-25 mg, chlorthalidone 12.5-25 mg, indapamide 1.25-2.5 mg): reduce sodium and water retention. Chlorthalidone is preferred due to longer duration of action. Monitor for hypokalemia, hyperuricemia, and glucose intolerance.

Second-Line and Special Situations

Beta-blockers (metoprolol succinate 25-200 mg, atenolol 25-100 mg, bisoprolol 2.5-10 mg): reduce heart rate and cardiac output. Particularly useful when hypertension is accompanied by angina, post-MI, heart failure, or tachyarrhythmia. Not first-line for uncomplicated hypertension.

Aldosterone antagonists (spironolactone 25-50 mg): effective add-on therapy, especially for resistant hypertension. Also beneficial in heart failure. Monitor potassium closely — risk of hyperkalemia, especially with ACE inhibitors or ARBs.

Alpha-blockers (doxazosin 1-8 mg): useful when hypertension accompanies benign prostatic hypertrophy. Central alpha-2 agonists (clonidine, methyldopa) — methyldopa is the drug of choice in hypertension during pregnancy.

Resistant hypertension (BP >140/90 on 3 medications including a diuretic) affects about 10% of hypertensive patients. Requires full-dose triple therapy, medication adherence assessment, ruling out secondary causes, and possible specialist referral.

8. Hypertensive Emergency — Emergency Protocol

Hypertensive emergency requires immediate controlled blood pressure reduction in a monitored setting (ICU/HDU). The goal is NOT to normalize BP immediately, as rapid

reduction can cause ischemia (watershed stroke, myocardial ischemia, acute tubular necrosis).

Target: reduce mean arterial pressure (MAP) by no more than 20-25% in the first hour, then to 160/100-110 over 2-6 hours, and to normal over 24-48 hours (except aortic dissection where faster reduction is needed).

IV medications used: labetalol (bolus 20 mg IV then infusion), nicardipine infusion, sodium nitroprusside infusion (ICU only, cyanide toxicity risk), hydralazine IV (preferred in eclampsia), esmolol infusion (aortic dissection).

Oral nifedipine (immediate release) is no longer recommended for hypertensive emergencies due to unpredictable BP drops. Sublingual captopril may be used for urgency (not emergency).

Specific situations: aortic dissection — target SBP <120 with heart rate control using IV beta-blocker + vasodilator. Eclampsia — IV labetalol or hydralazine, magnesium sulfate for seizure prevention. Hypertensive encephalopathy — reduce BP gradually, avoid overshooting.

9. Special Populations

Elderly patients (>65 years): Start low, go slow. Start with half the usual dose. Target SBP 130-139 in fit elderly, 140-149 in frail elderly. CCBs and diuretics are preferred. Avoid orthostatic hypotension — always check standing BP.

Diabetes and hypertension: ACE inhibitors or ARBs are preferred for their renal protective effects (reduce proteinuria). Target BP <130/80 mmHg. Avoid beta-blockers as first-line (can mask hypoglycemia symptoms).

Chronic Kidney Disease: ACE inhibitors or ARBs are first-line regardless of diabetes status. Monitor creatinine and potassium closely. Loop diuretics (furosemide) preferred over thiazides when GFR <30.

Pregnancy hypertension: Safe drugs include methyldopa, nifedipine, labetalol, and hydralazine. ACE inhibitors, ARBs, and direct renin inhibitors are absolutely contraindicated in all trimesters.

Young patients and women of childbearing age: Evaluate for secondary causes. CCBs and beta-blockers are often preferred. Thiazides can affect fertility.

10. Long-Term Management and Prognosis

Hypertension requires lifelong management. Uncontrolled hypertension significantly increases the risk of: stroke (7x), heart failure (3-4x), coronary artery disease (3x), chronic kidney disease (5x), peripheral artery disease, and retinopathy.

With appropriate treatment, cardiovascular risk can be substantially reduced. Each 10 mmHg reduction in systolic BP reduces major cardiovascular events by ~20%, stroke by ~35%, and heart failure by ~40%.

Medication adherence is the single greatest challenge in hypertension management. Pill burden, side effects, cost, and lack of symptoms all contribute to poor adherence. Once-daily combination pills (polypill approach) improve adherence significantly.

Regular monitoring: BP at every clinic visit, renal function and electrolytes every 6-12 months, fasting lipids and glucose annually, ECG and urine albumin annually, fundoscopy every 1-2 years.

Patient education is essential: Understanding that hypertension has no symptoms, the importance of taking medications even when feeling well, DASH diet principles, home BP monitoring technique, and when to seek urgent care (BP >180/120 or any warning symptoms).